

Synthesis, Magnetic Properties, and Electronic Structure of Magnetic Topological Insulator MnBi_2Se_4

Tiancong Zhu,[▽] Alexander J. Bishop,[▽] Tong Zhou, Menglin Zhu, Dante J. O'Hara, Alexander A. Baker, Shuyu Cheng, Robert C. Walko, Jacob J. Repicky, Tao Liu, Jay A. Gupta, Chris M. Jozwiak, Eli Rotenberg, Jinwoo Hwang, Igor Žutić, and Roland K. Kawakami*



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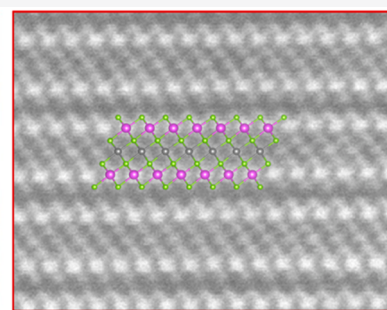
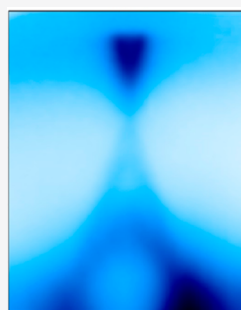


Supporting Information

ABSTRACT: The intrinsic magnetic topological insulators MnBi_2Te_4 and MnBi_2Se_4 support novel topological states related to symmetry breaking by magnetic order. Unlike MnBi_2Te_4 , the study of MnBi_2Se_4 has been inhibited by the lack of bulk crystals, as the van der Waals (vdW) crystal is not the thermodynamic equilibrium phase. Here, we report the layer-by-layer synthesis of vdW MnBi_2Se_4 crystals using nonequilibrium molecular beam epitaxy. Atomic-resolution scanning transmission electron microscopy and scanning tunneling microscopy identify a well-ordered vdW crystal with septuple-layer base units. The magnetic properties agree with the predicted layered antiferromagnetic ordering but disagree with its predicted out-of-plane orientation.

Instead, our samples exhibit an easy-plane anisotropy, which is explained by including dipole–dipole interactions. Angle-resolved photoemission spectroscopy reveals the gapless Dirac-like surface state, which demonstrates that MnBi_2Se_4 is a topological insulator above the magnetic-ordering temperature. These studies show that MnBi_2Se_4 is a promising candidate for exploring rich topological phases of layered antiferromagnetic topological insulators.

KEYWORDS: Molecular beam epitaxy, magnetism, topological insulator, angle-resolved photoemission spectroscopy



1. INTRODUCTION

The realization of novel topological quantum states is strongly correlated to the symmetries of different material systems.^{1–3} In topological insulators (TIs), the existence of time reversal symmetry (TRS) yields a topologically protected gapless surface state distinguished from its gapped bulk states.^{4–7} Breaking the TRS can lead to the quantum anomalous Hall (QAH) effect with dissipationless chiral edge states,^{8–12} the axion insulator states with quantized magnetoelectric effects,^{13–15} and Majorana fermions (by coupling to superconductors) obeying non-Abelian statistics.^{16–18} Experimentally, the engineering of TRS in TIs has been achieved through doping the TIs with dilute transition metals (Cr, Mn, V, etc.),¹⁹ where both QAH^{20–22} and axion insulator states^{23,24} have been observed at low temperature. However, one drawback of introducing magnetism through dilute doping is that the randomly distributed magnetic dopants can also induce disorder and nonuniformity in the sample, which ultimately limits QAH and axion insulator states from being realized at elevated temperatures.^{25,26}

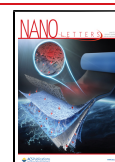
Consequently, researchers have increasingly focused on exploring intrinsic magnetic topological insulators, where the magnetic elements are ordered within the crystal lattice.¹⁴ Among such systems, MnBi_2X_4 ($\text{X} = \text{Se}, \text{Te}$) has garnered substantial attention due to predictions of novel topological

phases related to layered antiferromagnetic order.^{27–31} These materials have a layered van der Waals (vdW) structure with covalently bonded septuple layers (SLs) ($\text{X}-\text{Bi}-\text{X}-\text{Mn}-\text{X}-\text{Bi}-\text{X}$) as the base unit (Figure 1a): essentially the magnetic $\text{Mn}-\text{X}$ layer is inserted into each $\text{X}-\text{Bi}-\text{X}-\text{Bi}-\text{X}$ layer of the corresponding TI. Although there have been experimental reports of axion insulator and QAH insulator phases in MnBi_2Te_4 ,^{32,33} progress on topological phases in the selenide-based MnBi_2Se_4 has been limited by material synthesis. Efforts to synthesize bulk crystals have produced monoclinic MnBi_2Se_4 composed of magnetic chains,^{34,35} its thermodynamically stable phase, as opposed to the desired vdW crystal. In addition, thin film growth has only produced isolated SLs within Bi_2Se_3 ³⁶ or at its surface,^{37,38} while the synthesis of multilayer vdW MnBi_2Se_4 crystals needed for investigating interlayer magnetic coupling and topological effects has remained elusive.

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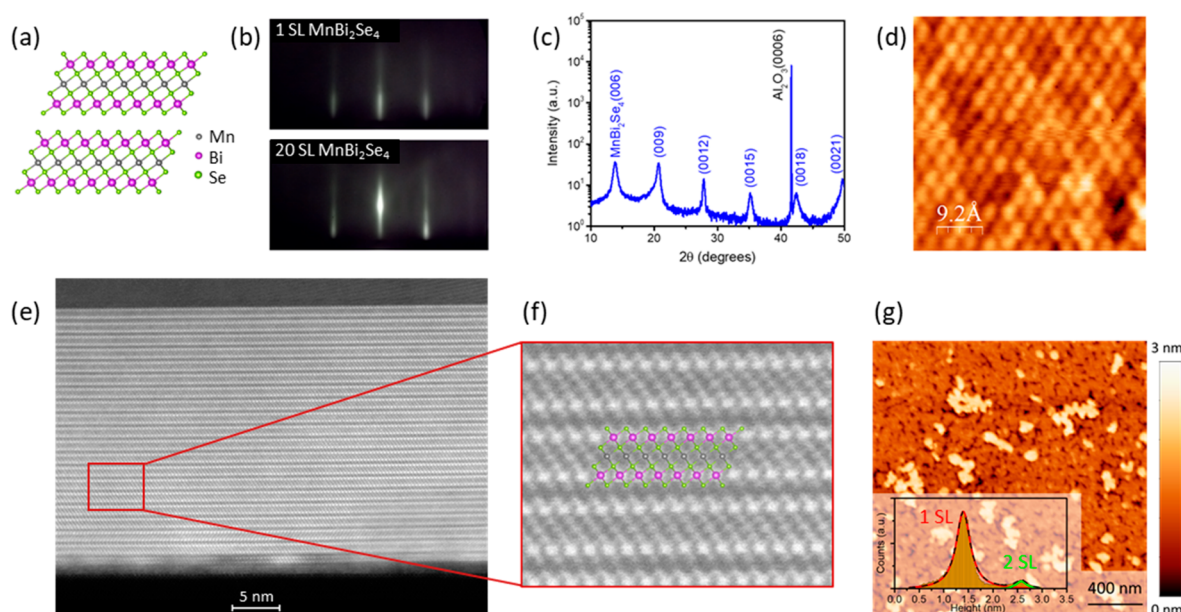


Figure 1. Structural characterization of MnBi_2Se_4 on $\text{Al}_2\text{O}_3(0001)$. (a) The atomic lattice structure of multilayer MnBi_2Se_4 consists of Se–Bi–Se–Mn–Se–Bi–Se septuple layers (SLs) in a van der Waals stack. (b) Streaky RHEED patterns on MnBi_2Se_4 films indicate a two-dimensional surface structure at thicknesses of 1 SL and 20 SL. (c) θ – 2θ X-ray diffraction scans of a 20 SL MnBi_2Se_4 show that only the substrate ($\text{Al}_2\text{O}_3(0001)$) and the $\text{MnBi}_2\text{Se}_4(00l)$ planes are observed. (d) STM imaging reveals the atomic-scale surface structure of a 20 SL MnBi_2Se_4 film with a hexagonal surface structure and an in-plane lattice constant of 4.0 ± 0.1 Å. (e) Cross-sectional STEM imaging reveals the van der Waals stacking and atomic-scale structure of the film with the substrate on the bottom and an Se cap on the top. (f) Enlarged STEM image with atomic lattice overlay that matches with MnBi_2Se_4 . (g) AFM images of a 1 SL film indicate a relatively flat film with regions of exposed substrate and the onset of second-layer nucleation. The height histogram (inset) shows the predominance of the monolayer film. The red and green solid lines show the Lorentzian fitting of the 1 SL and 2 SL profiles in the histogram, respectively, and the black dashed line shows the total fitting.

Here, we have successfully synthesized multilayer MnBi_2Se_4 vdW crystal films (up to 20 layers) using nonequilibrium molecular beam epitaxy (MBE) to stabilize the vdW layers and reveal their promising magnetic and topological properties. The atomic-scale structures have been verified by scanning transmission electron microscopy (STEM), scanning tunneling microscopy (STM), and atomic force microscopy (AFM). The magnetic properties have been investigated by superconducting quantum interference device (SQUID) magnetometry. A systematic study involving 10 samples of varying thicknesses reveal magnetic properties in agreement with the predicted layered antiferromagnetic ordering. On the other hand, the magnetic moments are found to lie in-plane, which disagrees with prior density functional theory (DFT) calculations predicting perpendicular magnetocrystalline anisotropy.^{27,31,37} We reconcile this discrepancy by including the magnetic shape anisotropy (i.e., dipole–dipole interaction) in the magnetic anisotropy energy calculations, which leads to an overall magnetic anisotropy favoring in-plane orientation of the moments. The electronic band structure of MnBi_2Se_4 has been investigated by angle-resolved photoemission spectroscopy (ARPES) and DFT calculations. ARPES measurements of the paramagnetic phase reveal the presence of topological surface states with Dirac dispersion that lie within the bulk band gap, which demonstrates that MnBi_2Se_4 is a TI above the magnetic ordering temperature. Our initial study shows the promise of multilayer MnBi_2Se_4 for exploring the rich topological phases of layered antiferromagnetic TIs. These results also open new avenues of research into vdW heterostructures with the potential for achieving tunneling planar Hall effects, magnetic proximity effects,^{39,40} and synthetic spin–orbit coupling.^{41–46}

2. SYNTHESIS AND STRUCTURAL CHARACTERIZATION

Epitaxial MnBi_2Se_4 films are grown on $\text{Al}_2\text{O}_3(0001)$ substrates in a Veeco 930 MBE system (see section 1 in the Supporting Information for details). With the substrate held at 275°C and under Se flux, we grow layers of MnBi_2Se_4 following a three-step process for each SL.⁴⁷ First, a quintuple layer of Bi_2Se_3 is grown by depositing the equivalent of two atomic layers of Bi. Next, we deposit a single atomic layer of Mn under Se flux and then wait for 5 min to allow the Mn to diffuse into the underlying Bi_2Se_3 template layer to form an individual trigonal MnBi_2Se_4 SL (Figure 1a). The reflection high energy electron diffraction (RHEED) images after each step can be found in Figures S2 and S3 of section 2 in the Supporting Information. This three-step process is repeated to create multilayer MnBi_2Se_4 to the desired thickness up to 20 SL, where sharp and streaky RHEED patterns persist throughout the growth (Figure 1b). The samples are typically capped with ~ 10 nm amorphous Se before being removed from the MBE chamber for further characterization.

Various structural characterizations of the samples confirm the good crystalline quality of the material. θ – 2θ X-ray diffraction measurements on a 20 SL sample show a series of peaks that correspond to the $\text{MnBi}_2\text{Se}_4(00l)$ planes (Figure 1c). These peaks yield an interplanar spacing of 12.76 Å in the sample, which is distinct from those of Bi_2Se_3 or other Mn–Se compounds. STM measurements on a 20 SL MnBi_2Se_4 sample (Figure 1d) reveal the hexagonal atomic structure of the sample surface consistent with imaging of the topmost Se layer, with an in-plane lattice constant of 4.0 ± 0.1 Å. Additional STM images are shown in section 3 in the Supporting Information. The crystal structure of the MBE-grown MnBi_2Se_4 is further investigated with cross-sectional STEM (see section 1 in the

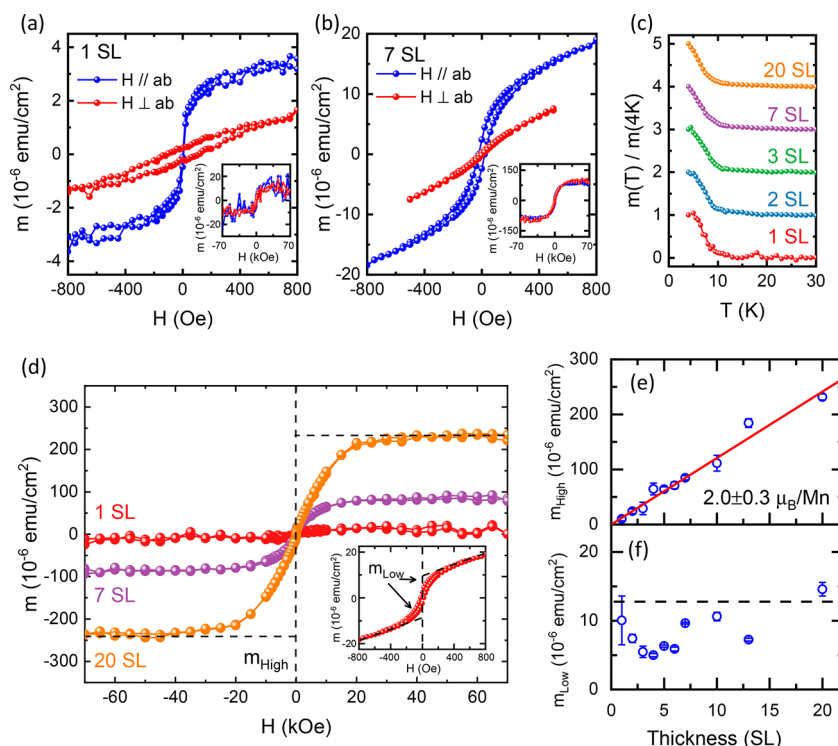


Figure 2. Magnetic properties of MnBi₂Se₄. (a) SQUID magnetization loops of a 1 SL MnBi₂Se₄ film at 4 K indicate ferromagnetic ordering with magnetic anisotropy favoring in-plane magnetization. The inset shows magnetization loops with higher field. (b) SQUID magnetization loops of a 7 SL MnBi₂Se₄ film at 4 K indicate ferromagnetic ordering with magnetic anisotropy favoring in-plane magnetization. The inset shows magnetization loops with higher field. (c) M vs T curves of MnBi₂Se₄ with different thicknesses shows a T_C value of ~ 10 K. The curves are measured with a small applied field of 50 Oe. (d) A comparison of in-plane magnetization loops of 1, 7, and 20 SL films shows an increase in the saturation moment per area (m_{High}) with increasing thickness. All loops exhibit low-field hysteresis, where we define the extrapolated remanence (m_{Low}) as shown in the inset. (e) A linear dependence of m_{High} with film thickness is consistent with aligned magnetic moments in high field. (f) m_{Low} showing little variation with film thickness is consistent with a low-field antiferromagnetic state with net uncompensated moments or surface ferromagnetism. The dashed line marks the magnetic moment for 1 SL MnBi₂Se₄ based on $2.0 \mu_B/\text{Mn}$ as determined by the slope in (e). Note: m_{Low} for the 1 SL sample in (f) is determined using the saturation value at high magnetic field (same method as m_{High}), due to no obvious hysteresis loop being observed at low field (a).

Supporting Information). The high-angle annular dark field (HAADF) image in Figure 1e shows the vdW stacking of SLs of the sample with a quintuple layer (QL) forming at the substrate (bottom), which might be associated with substrate preparation by Bi₂Se₃ growth and desorption⁴⁸ (see section 1 in the Supporting Information). The atomic structure of each SL in the enlarged image (Figure 1f) agrees very well with the trigonal MnBi₂Se₄ structure. The STEM measurement also confirms the continuous growth of multilayer MnBi₂Se₄ SLs, where previous studies have only shown isolated MnBi₂Se₄ layers within Bi₂Se₃ or at its surface.^{36,37} Figure 1g shows the atomic force microscopy (AFM) measurements on a nominally single layer MnBi₂Se₄ sample (this sample is not capped with Se). The vast majority of the film is composed of monolayer regions, and a second layer has started to nucleate in some areas (bright regions). The height difference between the two regions is 11.8 ± 1.0 Å, which agrees well with the step height of MnBi₂Se₄. The AFM measurement on the 1 SL sample also confirms that the growth of MnBi₂Se₄ is dominated by the layer-by-layer growth mode, as opposed to three-dimensional islands, which is important for interpreting the magnetic properties in section 3. These structural characterizations confirm the high quality of the MBE-grown MnBi₂Se₄ films.

3. MAGNETIC PROPERTIES

A systematic study of magnetic properties with thicknesses varying from 1 to 20 SL (Figure 2) provides evidence for easy-

plane magnetic anisotropy and layered antiferromagnetic order. Figure 2a shows the in-plane (blue) and out-of-plane (red) magnetization loops measured by SQUID at 4 K for a typical 1 SL sample. The in-plane loops exhibit a nonlinear ferromagnetic shape, while the out-of-plane loops are much flatter. These are signatures of ferromagnetism in the 1 SL sample with weak in-plane magnetic anisotropy. X-ray magnetic circular dichroism measurement on 1 SL samples further confirms that the magnetic signal originates from Mn atoms in our samples (section 4 in the Supporting Information). Similar ferromagnetic behavior is also observed for the 7 SL sample, as shown in Figure 2b, with a clearer hysteresis loop observed with in-plane measurement. Figure 2c shows the temperature dependence of the magnetization with a small in-plane field of 50 Oe for representative thicknesses (1, 2, 3, 7, 20 SL). With 4 K as the starting point, the magnetization decreases with increasing temperature and disappears at ~ 10 K for all thicknesses. Figure 2d shows representative in-plane hysteresis loops for 1, 7, and 20 SL film thicknesses. The multilayer samples exhibit a ferromagnetic-like hysteresis at low fields (inset), as we discussed in Figure 2b. The associated magnetic moment, denoted m_{Low} (total moment per sample area), is extrapolated from M vs H curves where the hysteresis loop is closed, as indicated in the inset of Figure 2d. On the other hand, even though the hysteresis loop is closed above ~ 500 Oe for the few-layer samples (e.g., Figure 2b), the measured magnetization does not saturate there. At higher fields, the magnetization gradually

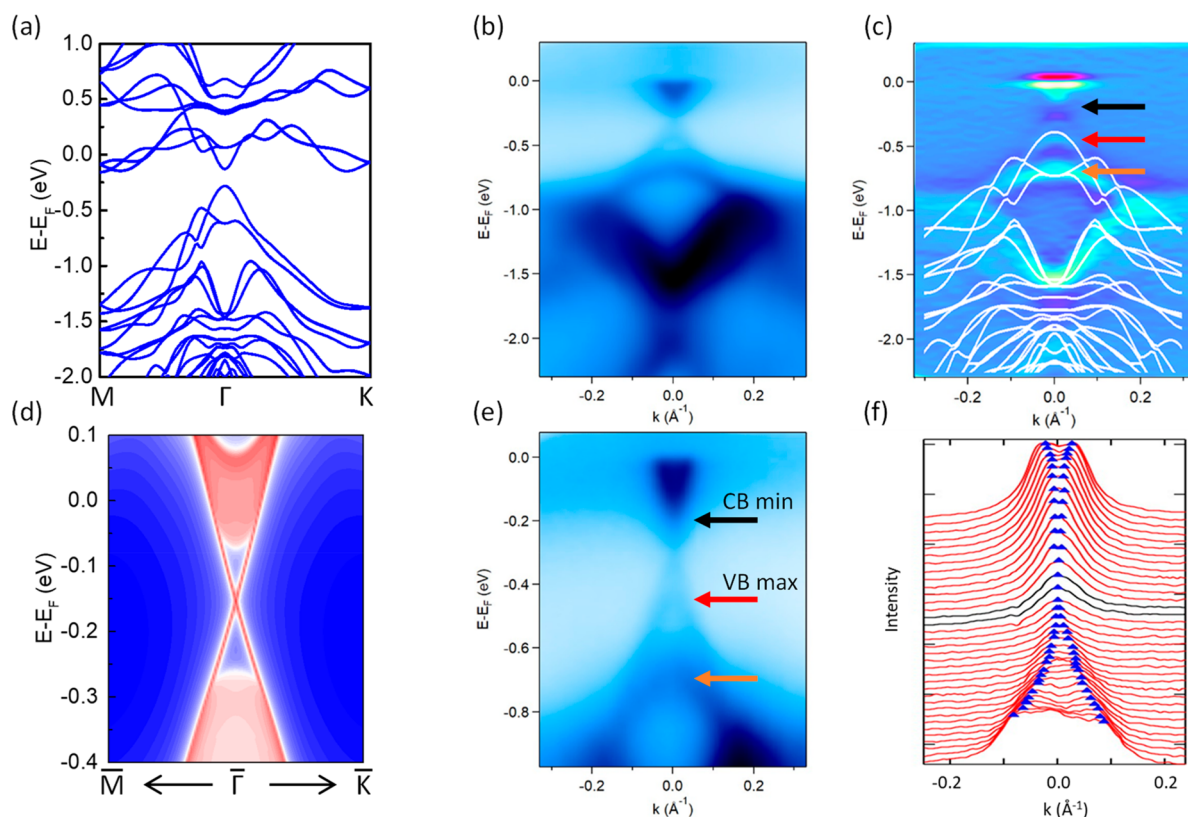


Figure 3. Electronic structure of MnBi_2Se_4 . (a) DFT calculation of the bulk states with Mn treated as nonmagnetic. (b) ARPES measurement of a 7 SL MnBi_2Se_4 film along $\text{M}-\Gamma-\text{M}$ with 20 eV photon energy and a sample temperature of 80 K. (c) An overlay of the DFT calculation and second derivative of the ARPES data in (b) shows a good agreement of calculation with the experiment. (d) DFT-calculated (001) surface states of nonmagnetic MnBi_2Se_4 . (e) Enlarged view of the ARPES data that highlights the Dirac surface state. (f) A series of momentum distribution curves show the evolution of the Dirac cone for evenly spaced energies ranging from -0.75 eV (bottom curve) to 0.0 eV (top curve). Local maxima are indicated by blue triangles. Black curves indicate the Dirac point. For (c) and (e), the black arrows indicate the conduction band minimum, red arrows indicate the valence band maximum (faint band), and orange arrows indicate a lower valence band (prominent band).

increases until it finally saturates and the saturation magnetic moment is denoted as m_{High} (total moment per sample area). The values of m_{High} and m_{Low} across 10 samples of varying thicknesses are summarized in Figures 2e,f, respectively. The magnitude of m_{High} shows a linear dependence on film thickness, where the slope indicates a magnetic moment of $2.0 \mu_{\text{B}}$ per Mn (Figure 2e). This is lower than the expected value of $5 \mu_{\text{B}}$ /Mn for Mn^{2+} ($S = 5/2$). While the origin of the reduced moment is unclear, prior studies of reduced magnetic moments in MnBi_2Te_4 , MnBi_4Te_7 , and $\text{MnBi}_{2-x}\text{Sb}_x\text{Te}_4$ provide insight into this issue^{49–56} (section 5 in the Supporting Information). In contrast to m_{High} , m_{Low} does not increase linearly with thickness and maintains a value of less than a monolayer of Mn (indicated by the dashed curve). Finally, transport measurements with features similar to those of the magnetic properties provide support that magnetic signals originate from the deposited films (section 6 in the Supporting Information).

The magnetic measurements are in agreement with DFT calculations^{27,31} for MnBi_2Se_4 that predict intralayer ferromagnetic coupling and interlayer antiferromagnetic coupling for a layered antiferromagnetic state. The intralayer ferromagnetic exchange interaction is established by our magnetic measurement for the 1 SL sample that exhibits ferromagnetic behavior (Figure 2a). To address the interlayer antiferromagnetic coupling, we discuss the magnetism in few-layer MnBi_2Se_4 . Although a ferromagnetic behavior is observed in few SL samples (Figure 2b), the magnetic moment associated with

ferromagnetism (m_{Low}) is significantly different from the total saturation moment of the samples (m_{High}). This shows that few-SL samples are not simply ferromagnetic. When the scaling of m_{High} and m_{Low} is compared with thickness (Figure 2e,f), the saturation moment m_{High} increases linearly with thickness, while m_{Low} does not vary much with thickness. This further rules out the scenario where the ferromagnetic signal is from random doping of Mn atoms, in which case m_{Low} should also scale with thickness (i.e., the amount of Mn deposition). Finally, by fitting the data in Figure 2e, we find the slope of $12.7 \pm 1.9 \text{ emu/cm}^2$ per SL corresponds to $2.0 \pm 0.3 \mu_{\text{B}}$ per Mn. Comparing this with the data in Figure 2f, we find that all of the m_{Low} values are within the magnetization of 1 SL MnBi_2Se_4 . All this evidence is consistent with layered antiferromagnetism with weak interlayer coupling. Under a low magnetic field, the Zeeman energy is weak in comparison to the interlayer AFM coupling; thus, the sample remains antiferromagnetic. However, due to the presence of monolayer thickness fluctuation (Figure 1g), certain regions of the few-layer MnBi_2Se_4 can have an odd number of layers where the magnetic moments are not compensated by antiferromagnetic ordering. As a result, a ferromagnetic-like feature with a total moment below the 1 SL value is observed. Under high magnetic field, the Zeeman energy can overcome the interlayer AFM coupling. In this case, all of the magnetic moments are aligned with the external field and, as a result, the total magnetic moment is scaled with film thickness.

The weak strength of interlayer AFM coupling is evidenced by the weak thickness dependence of the magnetic transition temperature. This can be understood in the mean-field picture. In a mean-field analysis, the ordering temperature T_C will be proportional to $z_F|J_F| + z_{AF}|J_{AF}|$, where z_F (z_{AF}) is the number of nearest neighbors and J_F (J_{AF}) is the ferromagnetic (antiferromagnetic) exchange coupling constant for moments in the same (neighboring) SL. In the limit of $|J_F| \gg |J_{AF}|$, the T_C value will be dominated by the ferromagnetic exchange. As a result, T_C should be insensitive to sample thickness, as is observed experimentally. While mean-field theory does not properly take dimensionality or fluctuations into account, it nevertheless shows how the two different exchange couplings contribute to a single magnetic phase transition and explains the negligible variation of T_C with thickness (Figure 2c) due to the dominance of J_F over J_{AF} .

Finally, we discuss the observed in-plane magnetic anisotropy in our MnBi_2Se_4 films. This is in apparent disagreement with previous DFT calculations of the magnetocrystalline anisotropy (MCA) in MnBi_2Se_4 that predict perpendicular magnetic anisotropy (PMA).^{27,31,37} To resolve this discrepancy, we calculated the magnetic anisotropy energy (MAE) including both the MCA and magnetic shape anisotropy (i.e., dipolar interactions) (see section 7 in the Supporting Information for details). Our calculations find that, while the MCA does exhibit PMA, its strength is sufficiently low that the inclusion of magnetic shape anisotropy yields an overall magnetic anisotropy favoring in-plane orientation of the moments.

4. ELECTRONIC STRUCTURE AND TOPOLOGICAL SURFACE STATES

We investigate the electronic and topological properties by performing ARPES measurements (see section 1 in the Supporting Information) on a 7 SL MnBi_2Se_4 film and compare them with DFT calculations. The ARPES measurements are performed at 80 K, where the sample is paramagnetically ordered. Simulating the paramagnetic order using DFT calculations is challenging. However, a nonmagnetic case can be a good approximation to capture the key electronic structure at the Γ point of the paramagnetic case, which has been verified in the similar system MnBi_2Te_4 .⁵⁷ Thus, we calculate the bands of a nonmagnetic MnBi_2Se_4 crystal as shown in Figure 3a. The key message here is the topological property of the bands at the Γ point. The calculated topological invariant Z_2 of 1 shows a nontrivial gap of ~ 0.2 eV at the Γ point, indicating that gapless Dirac surface states exist in the MnBi_2Se_4 surface and confirmed by the surface state calculation. To investigate the nature of these surface states, we perform DFT calculations on a semi-infinite slab of MnBi_2Se_4 . The calculation confirms the presence of a topological surface state with Dirac dispersion (red lines in Figure 3d) in addition to the bulk states, which appear as the solid red regions of Figure 3d in a projection onto the (001) plane.

Figure 3b shows the energy–momentum relation along the $M-\Gamma-M$ direction measured by ARPES (other cuts are shown in section 8 in the Supporting Information), which agrees well with the DFT calculation. To obtain a reliable identification of the bulk bands, we overlay the DFT calculation with the experimental data (Figure 3c), where we have taken the second derivative of the ARPES data along the energy axis to improve the contrast of the energy bands. The excellent overlap of the DFT calculation with the experimental spectra provides a confident assignment of the bulk conduction and valence bands.

On the basis of this assignment of bands, we identify the positions of the conduction band (CB) minimum (black arrow) and valence band (VB) maximum (red arrow) in the ARPES spectra.

The existence of a Dirac-like surface state in nonmagnetic MnBi_2Se_4 is clearly observed in enlarged ARPES spectra (Figure 3e), where the linearly dispersing bands are connecting the CB minimum (black arrow) and VB maximum (red arrow) of MnBi_2Se_4 . The bulk gap in ARPES (~ 0.25 eV in Figure 3e) is similar to that of the DFT calculation (~ 0.20 eV in Figure 3d). To investigate the surface state in the experimental data, we plot the momentum distribution curves for a series of energies that traverse the bulk band gap (Figure 3f). We observe a crossing of the local maxima (indicated by the blue triangles), which signifies the Dirac point and the gapless nature of the surface state within our energy resolution. Energy distribution curves (section 8 in the Supporting Information) further corroborate the gapless nature of the surface state.

5. DISCUSSION

It is worthwhile to compare our results to prior studies of related materials. Previous studies of Mn-doped Bi_2Se_3 reported in-plane ferromagnetism with $T_C \approx 10$ K, similar to our findings.^{58–60} Subsequent TEM studies of Mn-doped Bi_2Se_3 by Hagmann et al.³⁶ revealed the self-assembly of isolated MnBi_2Se_4 layers within Bi_2Se_3 , leading the authors to propose a re-evaluation of the origin of ferromagnetism in the previous studies. Our results further support this viewpoint, where the linear thickness dependence of the saturation moment (Figure 2e) and the direct visualization of MnBi_2Se_4 (Figure 1) indicate that our magnetic signals originate from MnBi_2Se_4 layers. In addition, we exclude the possibility of magnetic signals from dilute Mn doping of Bi_2Se_3 by growing directly on the Al_2O_3 substrates without a Bi_2Se_3 buffer layer. Thus, the similar magnetic ordering temperatures suggest MnBi_2Se_4 layers as a common origin of the magnetic signals.

Another study by Hirahara et al.³⁷ investigated the magnetic properties and band structure of a monolayer of MnBi_2Se_4 on Bi_2Se_3 . The results showed room-temperature ferromagnetism with perpendicular moments and the opening of a gap in the Bi_2Se_3 surface state. The magnetic properties are quite different from those of our current study but are similar to those of a recent study of epitaxial MnSe_2 films showing room-temperature ferromagnetism and perpendicular hysteresis.⁶¹ Further, the growth procedure developed in this paper is different from that of the earlier work, so it may be that the materials themselves are different. For instance, we have found in STM studies that the growth of MnSe_2 onto Bi_2Se_3 produces islands with different step heights and that these are sensitive to the growth conditions.⁶² A notable distinction of the present study is the ability to grow multilayer MnBi_2Se_4 as verified by STEM, which allows us to confidently ascribe the T_C value of ~ 10 K to multilayer MnBi_2Se_4 . Additional studies would be needed to reconcile the differences in the monolayer limit.

6. CONCLUSIONS

We have developed the MBE synthesis of multilayer trigonal MnBi_2Se_4 , a material that does not exist in bulk crystal form. STEM and STM imaging confirms the high quality of the material and identifies the atomic-scale structure of the septuple layer—the building block of the van der Waals crystal. Magnetic measurements indicate that the single-layer MnBi_2Se_4 is

ferromagnetic with a Curie temperature of ~ 10 K. The thickness-dependent magnetic properties agree with layered antiferromagnetic order. The observed magnetic properties, including the unexpected in-plane easy plane, are explained by DFT studies that include magnetocrystalline and magnetic shape anisotropy. These calculations along with ARPES measurements confirm the presence of a topological Dirac surface state, revealing that MnBi_2Se_4 in the paramagnetic state is a topological insulator. Our findings indicate the discovery of an intrinsically magnetic TI which could prove useful in future research endeavors.

With the realization of this high-quality magnetic TI we expect several interesting directions for future research in van der Waals heterostructures. In the normal state, magnetic TI/TI heterostructures could enable a peculiar tunneling planar Hall effect, which attains its maximum signal for the in-plane magnetization along the current direction, when the usual planar and other Hall effects vanish.^{41,42} In the superconducting heterostructures, such magnetic TIs could provide both magnetic proximity effects and synthetic spin–orbit coupling through magnetic textures to support spin-triplet superconductivity, an important element for superconducting spintronics^{43,44} and tunable Majorana bound states.^{42,45,46}

7. METHODS

See section 1 in the Supporting Information.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.1c00141>.

Methods (MBE, STEM, XMCD, ARPES, DFT), evolution of RHEED images during growth, additional STM images, XMCD data, discussion of the reduced magnetic moment, transport measurements, magnetic anisotropy calculations, and additional ARPES data (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Roland K. Kawakami – Department of Physics, The Ohio State University, Columbus, Ohio 43210, United States;
orcid.org/0000-0003-0245-9192; Email: kawakami.15@osu.edu

Authors

Tiancong Zhu – Department of Physics, The Ohio State University, Columbus, Ohio 43210, United States;
orcid.org/0000-0003-2703-5652
Alexander J. Bishop – Department of Physics, The Ohio State University, Columbus, Ohio 43210, United States
Tong Zhou – Department of Physics, University at Buffalo, Buffalo, New York 14260, United States
Menglin Zhu – Department of Materials Science and Engineering, The Ohio State University, Columbus, Ohio 43210, United States
Dante J. O'Hara – Department of Physics, The Ohio State University, Columbus, Ohio 43210, United States; Materials Science and Engineering, University of California, Riverside, California 92521, United States

Alexander A. Baker – Lawrence Livermore National Laboratory, Livermore, California 94550, United States;
orcid.org/0000-0003-0700-0858

Shuyu Cheng – Department of Physics, The Ohio State University, Columbus, Ohio 43210, United States

Robert C. Walko – Department of Physics, The Ohio State University, Columbus, Ohio 43210, United States

Jacob J. Repicky – Department of Physics, The Ohio State University, Columbus, Ohio 43210, United States

Tao Liu – Department of Physics, The Ohio State University, Columbus, Ohio 43210, United States

Jay A. Gupta – Department of Physics, The Ohio State University, Columbus, Ohio 43210, United States

Chris M. Jozwiak – Advanced Light Source, Lawrence Berkeley National Laboratory, Berkeley, California 94720, United States; orcid.org/0000-0002-0980-3753

Eli Rotenberg – Advanced Light Source, Lawrence Berkeley National Laboratory, Berkeley, California 94720, United States; orcid.org/0000-0002-3979-8844

Jinwoo Hwang – Department of Materials Science and Engineering, The Ohio State University, Columbus, Ohio 43210, United States

Igor Zutic – Department of Physics, University at Buffalo, Buffalo, New York 14260, United States

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.nanolett.1c00141>

Author Contributions

[†]T.Z. and A.J.B. contributed equally.

Author Contributions

R.K.K. and T.Zhu conceived the study. T.Zhu and A.J.B. synthesized the material with the help of D.J.O. T.Zhu performed the magnetometry measurement and data analysis with the help of A.J.B. and D.J.O. T.L. and T.Zhu performed the transport measurements. M.Z. performed the STEM measurement. T.Zhu and R.C.W. performed the STM measurement with the help of J.J.R. D.J.O. and A.A.B. performed the XMCD measurement. T.Zhu, A.J.B., S.C., and R.K.K. performed the ARPES measurements with C.M.J. and E.R. T.Zhou and I.Z. performed DFT calculations and theoretical analysis. All authors contributed to the interpretation of the results and preparation of the manuscript.

Notes

The authors declare no competing financial interest.

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Supporting Information

Synthesis, Magnetic Properties, and Electronic Structure of Magnetic Topological Insulator

MnBi₂Se₄

Tiancong Zhu,¹ Alexander J. Bishop,¹ Tong Zhou,² Menglin Zhu,³ Dante J. O'Hara,^{1,4} Alexander A. Baker,⁵ Shuyu Cheng,¹ Robert C. Walko,¹ Jacob J. Repicky,¹ Tao Liu,¹ Jay A. Gupta,¹ Chris M. Jozwiak,⁶ Eli Rotenberg,⁶ Jinwoo Hwang,³ Igor Žutić,² and Roland K. Kawakami¹

¹*Department of Physics, The Ohio State University, Columbus, OH 43210, USA*

²*Department of Physics, University at Buffalo, Buffalo, NY 14260, USA*

³*Department of Materials Science and Engineering, The Ohio State University, Columbus, OH 43210, USA*

⁴*Materials Science and Engineering, University of California, Riverside, CA 92521, USA*

⁵*Lawrence Livermore National Laboratory, Livermore, CA 94550, USA*

⁶*Advanced Light Source, Lawrence Berkeley National Laboratory, Berkeley, CA 94720, USA*

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1. Methods

Molecular beam epitaxy. Epitaxial growth is performed in a Veeco 930 MBE chamber. $\text{Al}_2\text{O}_3(0001)$ substrates (MTI Corporation) are annealed in air at 1000 °C for 3 hours then annealed under ultrahigh vacuum (UHV) at 600 °C for 30 minutes. Films are grown using elemental Mn (99.98%, Alfa Aesar) and Bi (99.998%, Alfa Aesar) deposited from effusion cells and Se (99.999%, Alfa Aesar) deposited from a valved cracker (900 °C in cracking zone, 290 °C in the bulk zone). Typical growth rates are 2.5 Å/min for Mn and 1.7 Å/min for Bi, while the Se flux is maintained at overpressure ($\sim 1 \times 10^{-6}$ torr) with flux ratios of ~ 100 for Se:Mn and Se:Bi as measured by the beam flux monitor, which is an ion gauge that measures the beam equivalent pressure. For each source, the beam flux monitor was calibrated by depositing elemental films and measuring their thickness by x-ray reflectometry. Similar to Levy *et al.*¹, we initiate the growth with a submonolayer amorphous Bi_2Se_3 layer at 140 °C, then re-evaporate the layer at 360 °C prior to the MnBi_2Se_4 film growth. This procedure helps to reduce the formation of twinning-domains and increases the surface smoothness of the MnBi_2Se_4 film. After growth, the sample is capped with 10 nm amorphous Se to preserve the sample as it is transferred to measurement systems.

Scanning transmission electron microscopy. Cross-section TEM specimens of MnBi_2Se_4 were made using focused ion beam (Thermofisher Helios), which were subsequently cleaned using a low energy ion mill (Fischione Nanomill) with the beam energy of 500 eV. STEM was performed using a Thermofisher Titan STEM operated at 300 kV. Atomic resolution high angle annular dark field (HAADF) images were acquired at the scattering range between 80 and 300 mrad, where the scattering intensity mostly depends on the Z-number of the atoms.

X-ray magnetic circular dichroism. The 1 SL film was capped with a thin (~ 3 nm) layer of amorphous Se to prevent oxidation and transported to beamline 4-ID-C of the Advanced Photon Source (APS). The sample was mounted and its surface contacted using silver paint. XMCD measurements were performed at the LHe cryostat's base temperature of 8 K, as measured by a thermocouple close to the sample. A magnetic field of 50 kOe was applied parallel to the beam, and x-rays were incident upon the sample at 30° from grazing incidence (60° from normal). X-ray absorption spectra (XAS) were gathered in surface-sensitive total electron yield detection, using positive and negative polarizations, and with positive and negative fields. Spectra were normalized to the incident beam intensity, measured with a gold mesh just upstream of the magnet chamber, and energy calibration was performed using a portion of the beam split off onto a MnO standard. A linear background was subtracted from the XAS spectra, and the pre-edge to post-edge step scaled to 1, to account for transitions into the continuum. XMCD was calculated as the difference between matched pairs (fixed polarization, switched field and switched polarization, fixed field) to ensure proper correction of incident beam effects.

Angle-resolved photoemission spectroscopy. The 7 SL MnBi_2Se_4 film is transported in an air-tight container under inert gas (N_2) to the Microscopic and Electronic Structure Observatory (MAESTRO) at the Advanced Light Source (ALS). The sample is loaded into UHV and given a mild anneal (~ 420 K) to remove the amorphous Se capping layer and then transferred into the microARPES endstation where it is cooled to ~ 80 K. The focused synchrotron beam has a spot size on the order of $10\ \mu\text{m}$ for photon energy of 20 eV used to obtain the data seen in figure 3. Photoemission spectra are collected using a Scienta R4000 electron analyzer equipped with custom-built deflectors that allow spectra over the first Brillouin zone to be taken on a single spot without moving the sample.

Density functional theory. The electronic band structures of MnBi_2Se_4 crystal were computed using the general potential linearized augmented plane-wave (LAPW) method² as implemented in the WIEN2K code³. Spin-orbit coupling is included as a second vibrational step using scalar-relativistic eigenfunctions as the basis after the initial calculation is converged to self-consistency. The lattice structure of MnBi_2Se_4 adopted in the calculations is shown in figure S1a. The optimized lattice constants of $a = b = 3.93\ \text{\AA}$ and $c = 36.78\ \text{\AA}$ are in a good agreement with our STM measurements. Figure S1b shows the high symmetry points of the Brillouin zone (BZ). The convergence of the calculations regarding the size of the basis set is achieved using an $R_{\text{MT}} * K_{\text{max}}$ value of 7, where R_{MT} is the smallest atomic sphere radius in the unit cell and K_{max} is the magnitude of the largest K wave vector inside the first BZ. The local density approximation (LDA) is used to describe the exchange and correlation functional. To take into account the correlation effects of Mn $3d$ electrons, the Hubbard U parameter is taken as 4 eV in the LDA+U calculations⁴. The Monkhorst-Pack k-grid of $12 \times 12 \times 3$ is adopted for the first BZ integral and the convergence criterion for the charge difference was less than $0.001e$ per unit cell. The surface states are calculated using the iterative Green's function technique as implemented in WannierTools package⁵ based on the tight-binding model constructed by the maximally localized Wannier functions⁶.

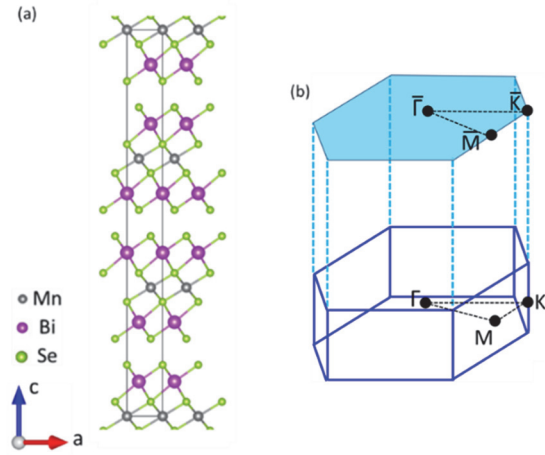


Figure S1. (a) The lattice structure of MnBi_2Se_4 crystal. The black lines indicate the unit cell adopted in the DFT calculations. (b) The bulk Brillouin zone (blue wireframe) and (001) surface Brillouin zone (shaded light blue). The high symmetry points used in Figure 3 of the main text are indicated here.

2. Evolution of RHEED images during MnBi_2Se_4 growth

The RHEED patterns seen in Figure 1b of the main paper were taken *in situ* with a standard RHEED system from STAIB Instruments. Figure S2 shows RHEED pattern images captured after each growth stage for the first three septuple layers (SLs). Each panel is labeled with a number and letter to indicate the growth layer and growth step respectively. As shown in the figure, letter “A” denotes an image taken immediately after the deposition of two atomic layers of Bi with the Se shutter open, called step A. Letter “B” labels images taken immediately after the deposition of one atomic layer of Mn also under Se flux, known as step B. Finally, letter “C” indicates images taken after the designed growth interrupt with Se flux, called step C.

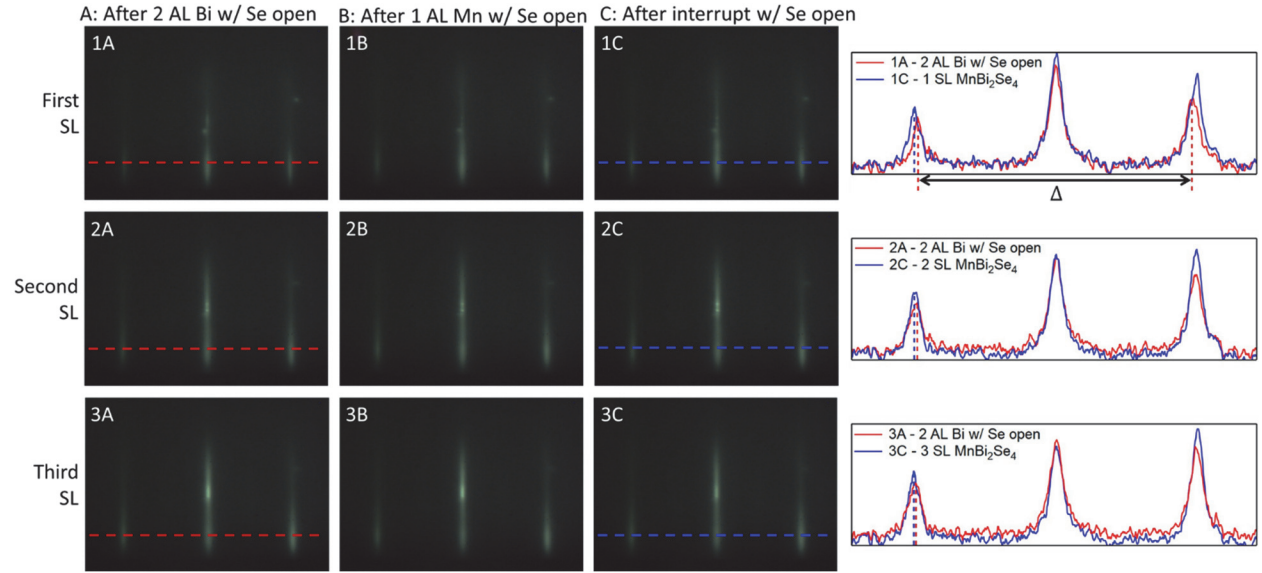


Figure S2. RHEED evolution of MnBi_2Se_4 growth. RHEED patterns taken along the $[11\bar{2}0]$ direction after each step of the growth process for the first three septuple layers of MnBi_2Se_4 growth. Each panel is given a label with the number indicating the septuple layer in progress and the letter indicating which step the growth process is at as described in the text. The graphs to the right of the image display intensity profiles for the first and last growth step of each septuple layer.

Intensity line cuts of RHEED images from step A and step C of each layer and can be seen on the right side of Figure S2. These profiles show that the spacing of the RHEED streaks, denoted by Δ , changes after the first and last steps of the growth. It can be seen that the Δ measured after the designed growth interrupt (step C) is larger than the Δ measured immediately after the deposition of Bi (step A) at each septuple layer. This Δ was measured after steps A and C for the first five SLs of a growth and analyzed in Figure S3. Each point in Figure S3 is a calculated percent difference between the initial 2 AL of Bi deposited for the first SL (labeled 1A in Figure S2) and the growth step being analyzed. Red dots indicate RHEED spacing measured after step A of a given SL growth while blue dots represent the spacing measured after the completion of a full SL. Note that for each SL, the Δ measured after step A is smaller than the Δ measured after step C. This indicates the formation of Bi_2Se_3 after each A step, and then a transition into MnBi_2Se_4 upon completion of

all growth stages. *In situ* analysis of this spacing is an early indication of successful growth of MnBi_2Se_4 SLs. Also note that the initial 2 AL of Bi (step 1A) compared to the spacing measured after the completion of the first SL (step 1C) gives a percent difference of $\sim 2.5\%$. This agrees well with the percent difference between the accepted value of the Bi_2Se_3 in-plane lattice constant of $4.14 \text{ \AA}$ and the measured MnBi_2Se_4 lattice constant of $4.0 \text{ \AA}$.

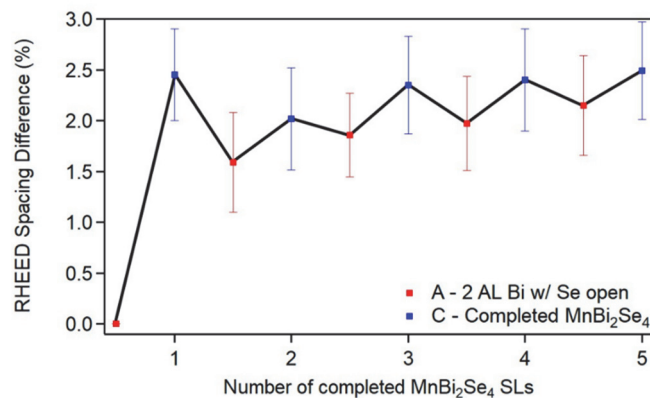


Figure S3. Evolution of RHEED spacing during growth. Percent difference of RHEED spacing at first and last steps of growth during the first five septuple layers.

3. Additional STM images

Figure S4 shows an STM image of a 20 SL MnBi_2Se_4 film taken over a $400 \times 400 \text{ nm}^2$ area. The tunneling conditions are -1.5 V sample bias and 0.2 nA tip current. The surface possesses smooth terraces with SL steps. Straight sections of step edges align along crystallographic axes. There are some similarities of the surface with $\text{Bi}_2\text{Se}_3(0001)$ which typically consist of triangular islands, but these MnBi_2Se_4 surfaces are qualitatively distinct.

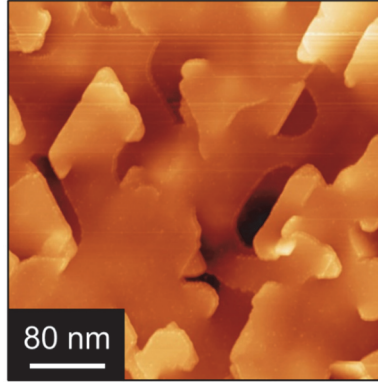


Figure S4. Large area STM image.

By adjusting the tunneling conditions to -1.0 V sample bias and 0.54 nA tip current, we observe subsurface Mn atoms on Bi sites (Mn_{Bi} antisites) as dark triangles (Figure S5), similar to those observed in Mn-doped Bi_2Te_3 ⁷ and MnBi_2Te_4 ⁸. We note that it is unclear whether this surface measurement represents the bulk properties. In MBE growth, it is often the case that dopant ions are pushed toward the surface due to surface band bending (SBB). In our case, if the SBB is similar to what has been measured for Bi_2Se_3 ⁹, then the relative force from the SBB will be to push Bi^{3+} into the bulk and Mn^{2+} toward the surface and therefore produce more Mn_{Bi} antisites near the surface than in the bulk. More studies would be needed to verify such a trend.

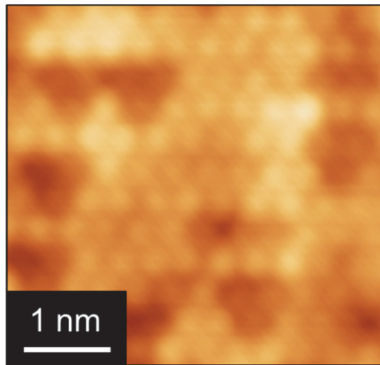


Figure S5. Imaging Mn_{Bi} antisite defects.

4. X-ray magnetic circular dichroism

The origin of the magnetic signal in single SL MnBi_2Se_4 is investigated by element-specific x-ray magnetic circular dichroism (XMCD) (Fig. S6). Circularly polarized x-rays are incident upon the sample at 30° from grazing, with magnetic field applied parallel to the x-ray beam. Detection is performed using surface-sensitive total electron yield mode. A difference between left- and right-circularly polarized x-ray absorption is measured at the Mn-L_{2,3} edges at 8 K and 50 kOe, demonstrating that the observed magnetic signal originates with Mn, ruling out the possibility of substrate impurities as the origin of the SQUID results.

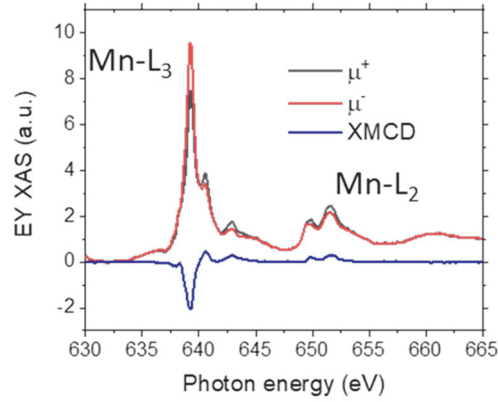


Figure S6. X-ray Magnetic Circular Dichroism data. Element-specific XMCD measurement of 1 SL MnBi_2Se_4 at 5 T and 8 K, where x-ray absorption spectra at the Mn L-edge for μ^+ circular polarization (black curve) and μ^- circular polarization (red curve) exhibits an asymmetry (blue curve) that confirms a net magnetic moment from Mn.

5. Discussion of the reduced magnetic moment

The thickness dependence of the saturation magnetic moment m_{high} in Figure 2e yields a magnetic moment of $2.0 \pm 0.3 \mu_B$ per Mn, which is lower than the expected value of $5 \mu_B/\text{Mn}$ for Mn^{2+} ($S = 5/2$). While the origin of the reduced moment observed experimentally is unclear, prior studies of reduced magnetic moments in related telluride materials MnBi_2Te_4 , MnBi_4Te_7 , and $\text{MnBi}_{2-x}\text{Sb}_x\text{Te}_4$ provide insight into this issue^{8, 10-16}.

Determining the magnetic moment from magnetic hysteresis loops has yielded values of $3.56 \mu_B/\text{Mn}$ ¹⁰ and $2.6 \mu_B/\text{Mn}$ (15 emu/g)¹¹ for MnBi_2Te_4 and $3.5 \mu_B/\text{Mn}$ ¹² and $\sim 3 \mu_B/\text{Mn}$ ¹³ for MnBi_4Te_7 , while determining the magnetic moment from neutron scattering has yielded values of $4.04 \mu_B/\text{Mn}$ ⁸ and $4.7 \mu_B/\text{Mn}$ ¹⁴ for MnBi_2Te_4 and $4.0 \mu_B/\text{Mn}$ ¹⁴ for MnBi_4Te_7 . This suggests a systematic underestimation of the magnetic moment by magnetization measurements compared to neutron scattering, where the former assumes a saturated state while the latter directly probes the magnetic order. Such effects could affect our magnetic moment values which are based on magnetization measurements.

It has also been suggested that reduced magnetic moment values could be related to magnetic fluctuations. One possible source of such fluctuations is magnetic frustration. In a study of $\text{MnBi}_{2-x}\text{Sb}_x\text{Te}_4$,¹⁰ a reduction of magnetic moment from $3.56 \mu_B/\text{Mn}$ for MnBi_2Te_4 to $1.67 \mu_B/\text{Mn}$ for MnSb_2Te_4 was observed. Due to a reduction in the Mn-Mn distance with Sb content, it was suggested that an enhanced direct antiferromagnetic interaction would compete with a dominant ferromagnetic interaction and produce magnetic fluctuations for a reduced moment in the ordered state. Another source of magnetic fluctuations is thermal energy. In comparing magnetic moments of MnBi_2Te_4 and MnBi_4Te_7 ,¹⁴ the neutron scattering values were $4.7 \mu_B/\text{Mn}$ and $4.0 \mu_B/\text{Mn}$, respectively. It was suggested that thermally-induced magnetic fluctuations at the measurement temperature of 4.5 K could produce greater fluctuations for MnBi_4Te_7 compared to MnBi_2Te_4 due to the lower ordering temperature (13 K for MnBi_4Te_7 vs. 23 K for MnBi_2Te_4), thus leading to a lower net magnetic moment. This is relevant for our MnBi_2Se_4 samples due to the even lower ordering temperature of 10 K.

Lastly, the presence of structural disorder such as Mn-related point defects could affect the net magnetic moment. Mn vacancies and Mn_{Bi} antisites (Mn on Bi sites) have been observed in MnBi_2Te_4 ^{8, 15, 16} and may have deleterious effects on the net magnetic moment. We have observed Mn_{Bi} defects in MnBi_2Se_4 (SI section 3).

Due to these various possible effects, further research is needed to understand the origin of reduced magnetic moment that we observe in MnBi_2Se_4 .

6. Transport measurements

To ensure that the SQUID magnetization signals originate from the MnBi_2Se_4 films as opposed to some artifacts (e.g. magnetic clusters, substrate, contamination), we performed transport measurements. Figure S7a shows the Hall resistance measured on a 20 SL MnBi_2Se_4 film measured at 4 K. The sample was grown on c-sapphire and capped with amorphous Te. After the growth, the sample was fabricated into a Hall bar geometry by scribing the unwanted region off with a diamond scribe. The field dependent resistance measurements were done in a PPMS at 4 K. The data has been antisymmetrized to remove contributions from longitudinal resistance R_{xx} due to imperfect sample geometry. The most notable feature of the data is the nonlinear shape of the curve as a function of out-of-plane field. This is similar to the out-of-plane SQUID magnetization loop which exhibits a nonlinear shape with saturation above ~ 30 kOe (inset of Fig. 2b, red curve). The origin of differences such as the reduction of Hall resistance above ~ 20 kOe are unclear and might be related to a topological Hall effect. Nevertheless, the overall similarity with the magnetization data provides support that the SQUID is measuring the magnetic properties of the MnBi_2Se_4 film as opposed to artifacts.

Further evidence that the SQUID measures the MnBi_2Se_4 film comes from the temperature dependence of R_{xx} (Fig. S7b). By plotting $\log(R_{xx})$ as a function of $T^{-1/3}$ (Fig. S7c), a linear dependence at lower temperatures indicates transport is by two-dimensional variable range hopping (VRH). Notably, there is a kink in the curve at ~ 10 K with a lower slope for $T > 10$ K (blue line) and a higher slope for $T < 10$ K (red line). Because the change in slope coincides with the magnetic ordering temperature $T_C \sim 10$ K (see Fig. 2c), this provides further support that the SQUID is measuring the magnetic properties of the MnBi_2Se_4 film.

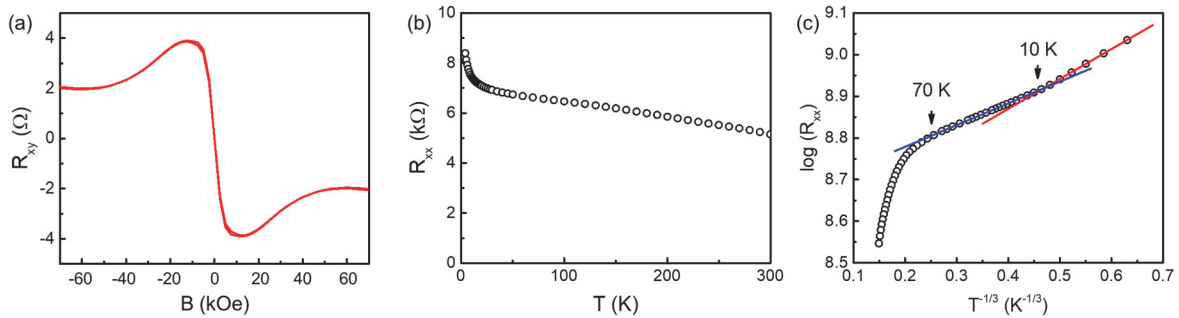


Figure S7. Transport measurements on 20 SL MnBi_2Se_4 . (a) Hall resistance at 4 K as a function of out-of-plane magnetic field. (b) Longitudinal resistance at zero magnetic field as a function of temperature. (c) Replotting the temperature dependence of longitudinal resistance to illustrate transport by two-dimensional variable range hopping. The blue and red lines show a change in slope at ~ 10 K.

7. Magnetic anisotropy calculation

The in-plane orientation of the magnetic moments is understood by calculating the magnetic anisotropy energy (MAE), which contains two main terms^{17, 18}: (1) magnetocrystalline anisotropy (MCA) energy (E_{MCA}) resulting from the spin-orbit coupling (SOC), and (2) magnetic shape anisotropy (MSA) energy (E_{MSA}) resulting from the dipole-dipole interaction; i.e., $MAE = E_{MCA} + E_{MSA}$. In calculations, E_{MCA} can be obtained by calculating the total energy difference as the magnetization rotates from the in-plane (*ip*) to the out-plane (*op*) direction ($E_{MCA} = E_{ip}^{SOC} - E_{op}^{SOC}$), while E_{MSA} ($E_{MSA} = E_{ip}^{dipole} - E_{op}^{dipole}$) can be calculated based on the energy of magnetic dipole-dipole interactions as^{17, 18}

$$E^{Dipole} = -\frac{1}{2} \frac{\mu_0}{4\pi} \sum_{i \neq j}^N \frac{1}{r_{ij}^3} \left[\vec{M}_i \cdot \vec{M}_j - \frac{3}{r_{ij}^2} (\vec{M}_i \cdot \vec{r}_{ij})(\vec{M}_j \cdot \vec{r}_{ij}) \right].$$

Here $\vec{M}_i$ represents the local magnetic moments and $\vec{r}_{ij}$ are vectors that connect the sites *i* and *j*.

The contribution from MSA is usually ignored in MAE calculations,¹⁹⁻²⁴ which is reasonable for a very strong SOC system.^{21-23, 25, 26} However, for some materials the MSA usually dominates the MAE,^{17, 18} which must be included. For 1 SL MnBi₂Se₄, our calculations show its $E_{MCA} = 0.04$ meV and $E_{MSA} = -0.12$ meV, indicating its MCA favors the out-plane magnetization while its MSA prefers in-plane magnetization. Their competitions determine the magnetic anisotropy. The calculated MAE of -0.08 meV indicates the 1 SL MnBi₂Se₄ has an in-plane magnetic anisotropy, consistent with our experimental observations. For comparison, we also carried out the MAE calculations for 1 SL MnBi₂Te₄ and found its calculated $E_{MCA} = 0.19$ meV and $E_{MSA} = -0.14$ meV, indicating MnBi₂Te₄ ML prefers out-plane magnetization with MAE of 0.05 meV, in agreement with the previous report^{21-23, 27} and experiments.^{28, 29} The E_{MSA} of MnBi₂Te₄ is similar to that of MnBi₂Se₄, but because of the much stronger SOC, the E_{MCA} of MnBi₂Te₄ is much larger than that in MnBi₂Se₄ and consequently dominates the MAE in MnBi₂Te₄. This is the reason why an in-plane magnetic anisotropy is observed in MnBi₂Se₄ system but an out-plane magnetic anisotropy in MnBi₂Te₄ system.

The competition between MCA and MSA may also produce a canting of magnetic moments at monolayer edges, which could be the origin of the weak hysteresis observed in the out-of-plane loop for the 1 SL sample (Fig. 2a, red curve). At the monolayer edges (see Fig. 1g), the reduced number of neighbors may weaken the MSA and allow the perpendicular MCA to produce a slight canting of the magnetic moments toward the out-of-plane direction.

8. Additional ARPES data

Figure S8 provides additional cuts of the ARPES data from a 7 SL sample. Figure S8a shows the ARPES measurement along the K- Γ -K direction. The black arrow indicates the conduction band minimum, the red arrow indicates the valence band maximum, and the orange arrow indicates a lower valence band. Two-dimensional momentum maps ($k_x - k_y$) are plotted for a series of binding energies, which highlight the momentum distribution of the states near the Γ -point (Figure S8b). The red dashed lines track the linear dispersion of the surface state. The gapless nature of the surface state is confirmed by investigating the energy distribution curves (EDC). Figure S8c shows the EDC at $k = 0.00 \text{ \AA}^{-1}$ (solid red curve) through the surface state and background EDCs at $k = 0.08 \text{ \AA}^{-1}$ (solid blue curve) and $k = -0.08 \text{ \AA}^{-1}$ (dashed blue curve) away from the surface state. For energies between -0.5 eV and -0.2 eV, the red curve remains well above the background curves indicating the absence of a gap in the Dirac surface state.

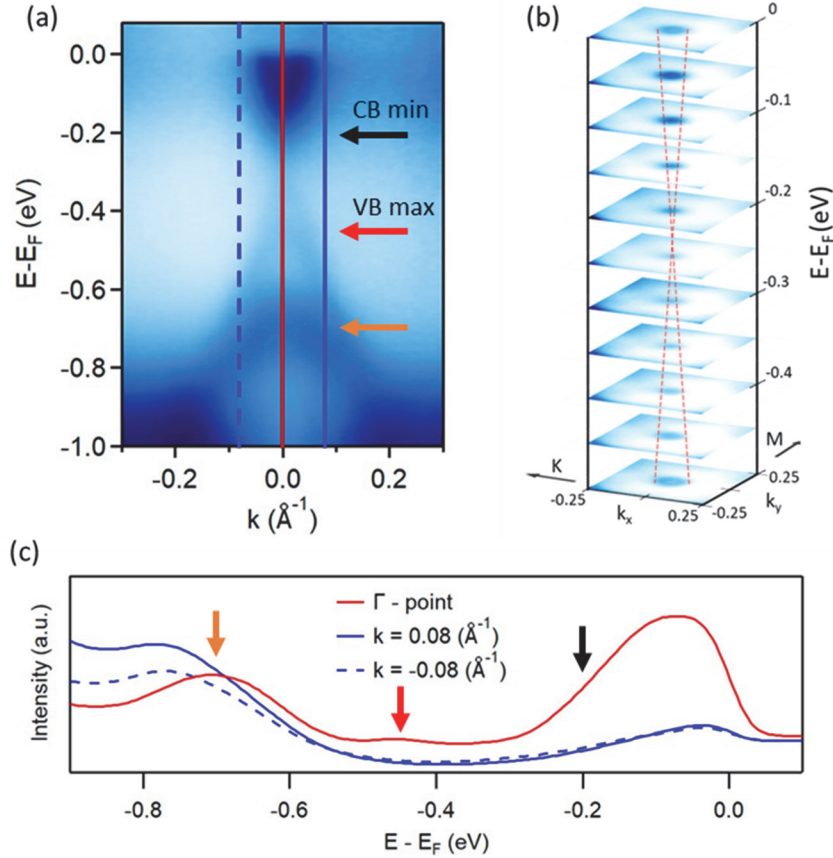


Figure S8. Additional ARPES data. (a) ARPES measurement of a 7 SL MnBi_2Se_4 film along the K- Γ -K direction. (b) Constant energy contours of the 7 SL film at different energies. (c) Energy distribution curves taken at $k = -0.08 \text{ \AA}^{-1}$ (dashed blue line), $0.08 \text{ \AA}^{-1}$ (solid blue line), and the gamma point (solid red line) as indicated in (a). For panels (a) and (c), the black arrows indicate the conduction band minimum, red arrows indicate the valence band maximum (faint band), and orange arrows indicate a lower valence band (prominent band). All measurements were taken with a photon energy of 20 eV and sample temperature of 80 K.

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